CEMAC Jobs Analytics

Cameroon elasticities and WBES trade diagnostics

September 30, 2026

Roadmap

1. BDF Cleaning Results
2. Cameroun Census data on firm counts, employment, and revenue
3. Cameroun Sectoral Elasticities
4. B-READY Mapped WBES Constraints
5. CEMAC WBES Trade Analysis

Data cleaning overview

- The working panel is organized at the firm-year level, so duplicate handling changes both sample size and identification.
- Source sectors are three-digit NACAM Rev.1 branches. The reviewed crosswalk assigns them to 16 categories based on the NACAM–ISIC Rev.4 crosswalk.
- Duplicate treatment and the documented section assignments determine the analysis sample and sector interpretation.

Firm-Year Duplicate Cleaning

- Working key: `firmid` × `fin_yr`
- Perfect duplicates would be collapsed with original row order as the tie-breaker.
- Conflicting duplicate firm-years are fully removed from the cleaned analysis file, while we figure out the best approach for handling them.

	Value
Original observations	9,325
Perfect duplicate firm-years	0
Rows dropped from perfect duplicates	0
Conflicting duplicate firm-years	127
Rows dropped from conflicting duplicates	257
Final cleaned observations	9,068

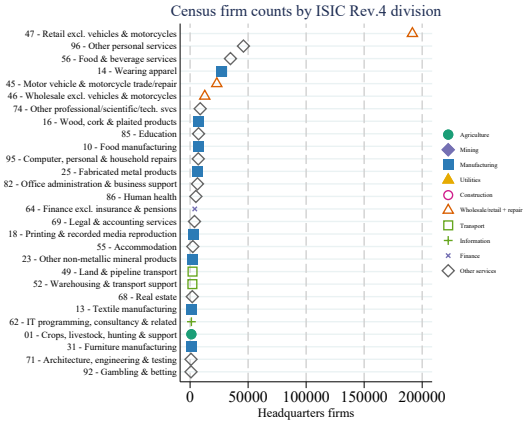
Cleaning Takeaways

- The cleaned panel resizes the sample from 9,325 to 9,068 observations after removing duplicate firm-years.
- There are 127 conflicting firm-years, accounting for 257 dropped obs. Those observations are excluded rather than resolved ad hoc.
- Ecofin uses ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. The combinations are professional & admin. support, and arts/recreation, other services & household employers (division 97). Crosswalk qualifications are listed in the appendix.

Cameroun Census Context

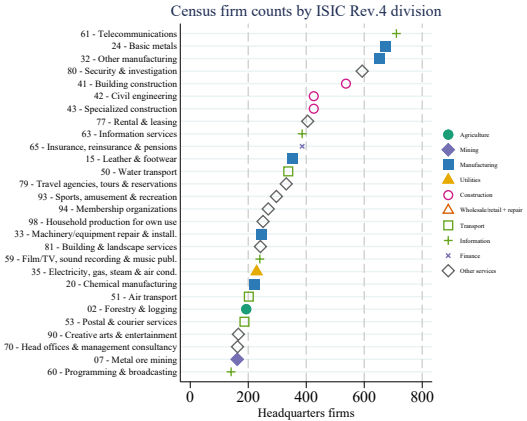
- The 2024 RGE census gives a whole-economy snapshot of establishments in Cameroon.
- Reviewed activity and firm-specific decisions assign Census firms to two-digit ISIC Rev.4 divisions. Unknown activities remain unclassified. Short English labels follow INSEE NAF Rev.2, which shares the same 88 divisions.
- The next slides use headquarter-level data to show sector scale in firm counts, log employment, log annual revenue, and log revenue per worker.
- After this economy-wide view, the administrative tax panel focuses on formal firms observed over time, using the administrative data.

Census Firm Counts by ISIC Rev.4 Division (1/3)



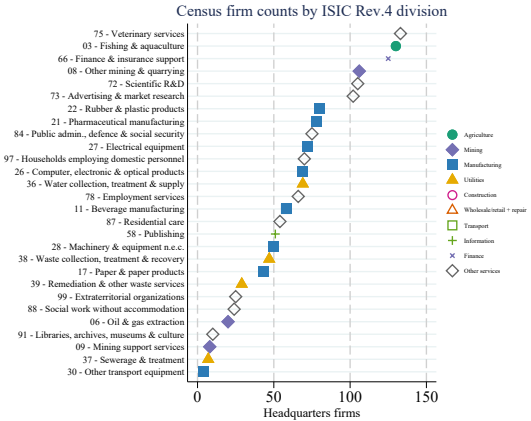
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Firm Counts by ISIC Rev.4 Division (2/3)



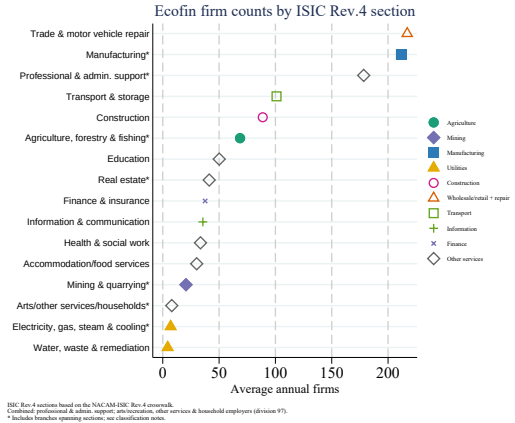
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Firm Counts by ISIC Rev.4 Division (3/3)



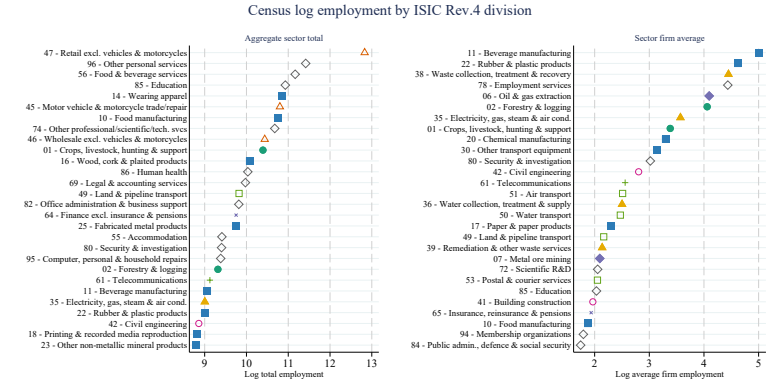
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Ecofin Firm Counts by ISIC Rev.4 Section



Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Log Employment by ISIC Rev.4 Division (1/3)

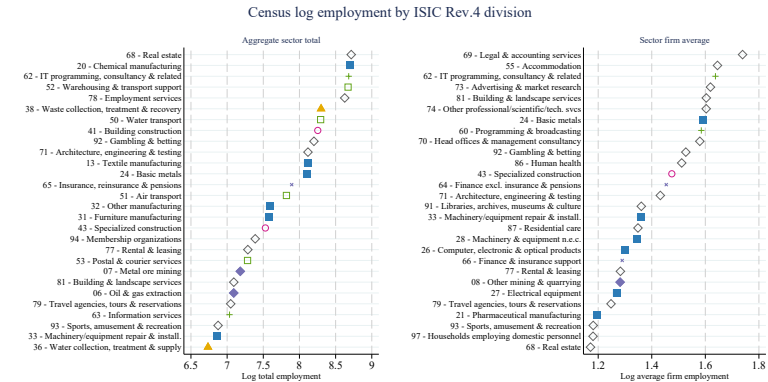


Ranks 1-28; each panel ranks divisions separately.

Appendix: same figure in levels

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Log Employment by ISIC Rev.4 Division (2/3)

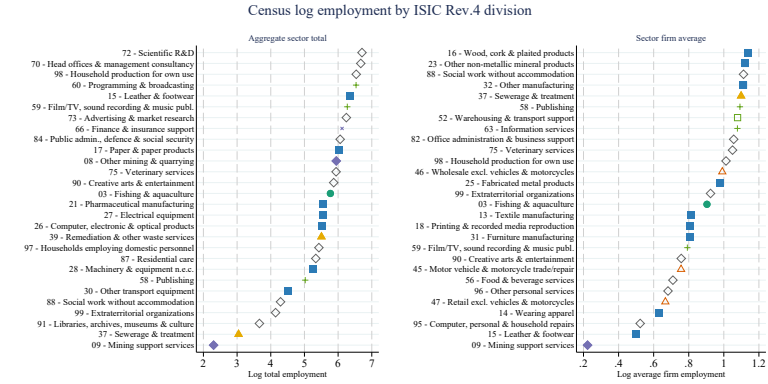


Ranks 29-56; each panel ranks divisions separately.

Appendix: same figure in levels

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Log Employment by ISIC Rev.4 Division (3/3)

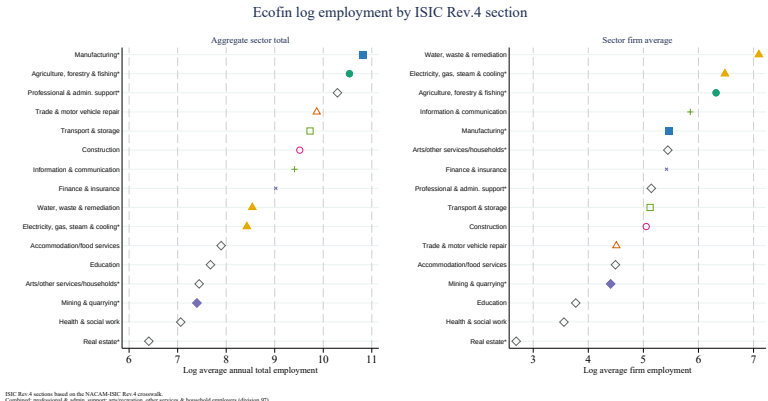


Ranks 57-84; each panel ranks divisions separately.

Appendix: same figure in levels

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

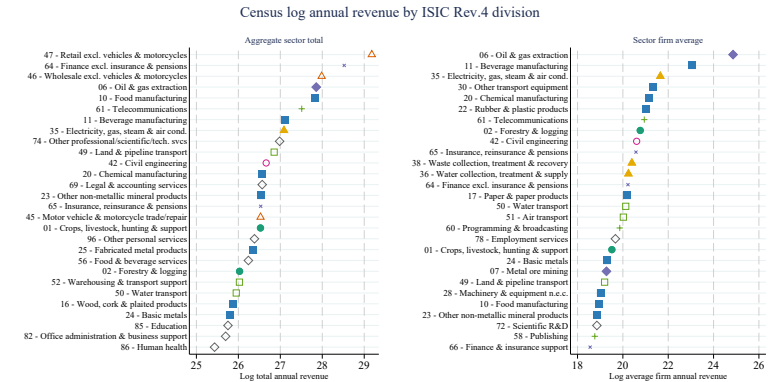
Ecofin Log Employment by ISIC Rev.4 Section



Appendix: same figure in levels

Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Log Revenue by ISIC Rev.4 Division (1/3)

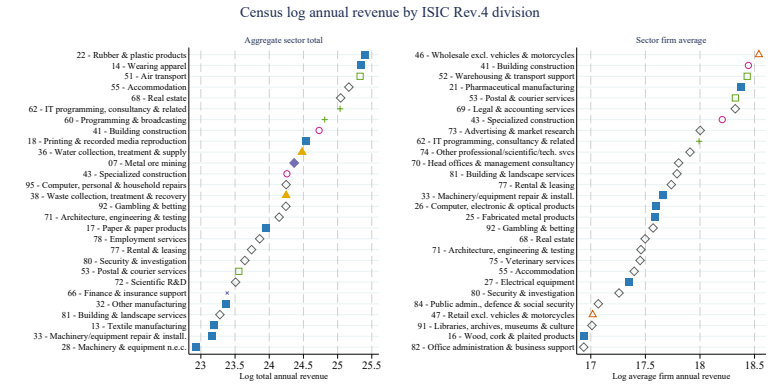


Ranks 1-28; each panel ranks divisions separately.

Appendix: same figure in levels

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Log Revenue by ISIC Rev.4 Division (2/3)

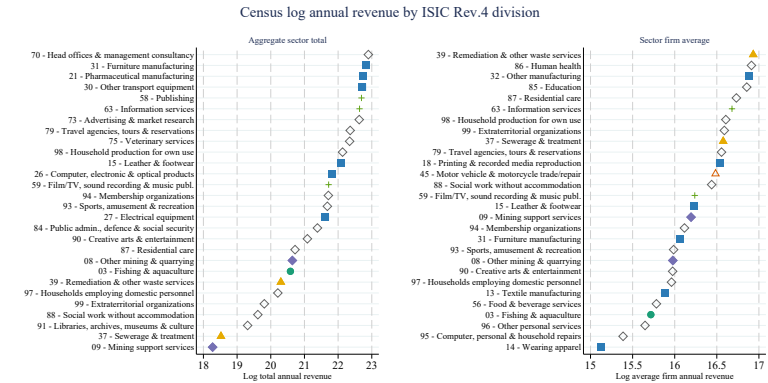


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Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Log Revenue by ISIC Rev.4 Division (3/3)

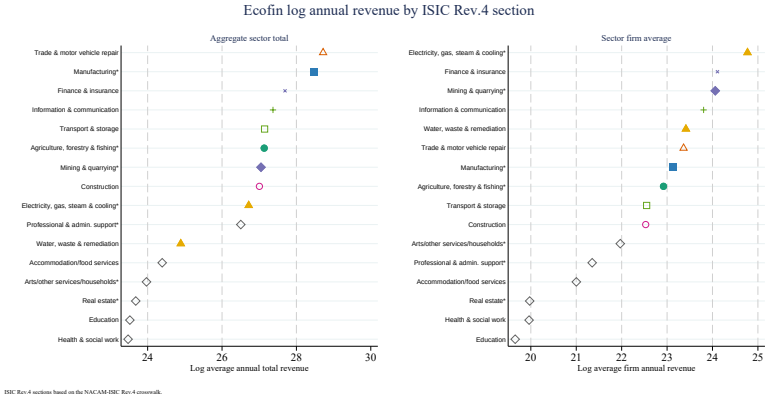


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Appendix: same figure in levels

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Ecofin Log Revenue by ISIC Rev.4 Section

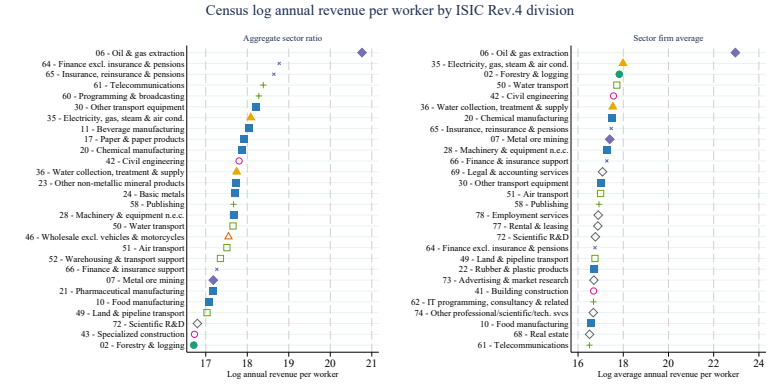


ISIC Rev.4 sections based on the NACAM-ISIC Rev.4 crosswalk.
Combined: professional & admin. support; arts/recreation, other services & household employers (division 97).
* Includes branches spanning sections; see classification notes.

Appendix: same figure in levels

Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Log Revenue per Worker by ISIC Rev.4 Division (1/3)

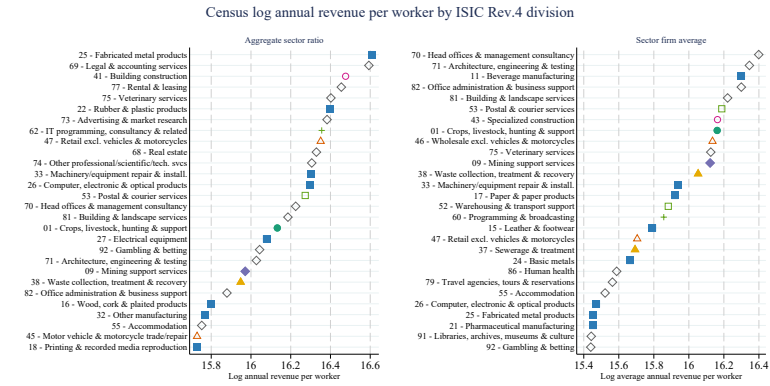


Ranks 1-28; each panel ranks divisions separately.

Appendix: same figure in levels

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Log Revenue per Worker by ISIC Rev.4 Division (2/3)

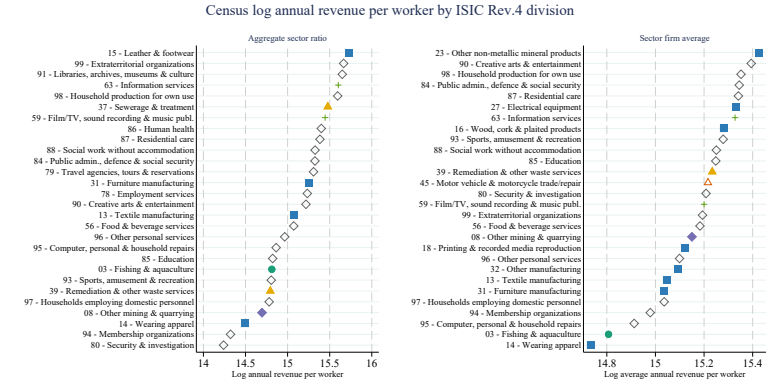


Ranks 29-56; each panel ranks divisions separately.

Appendix: same figure in levels

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Log Revenue per Worker by ISIC Rev.4 Division (3/3)

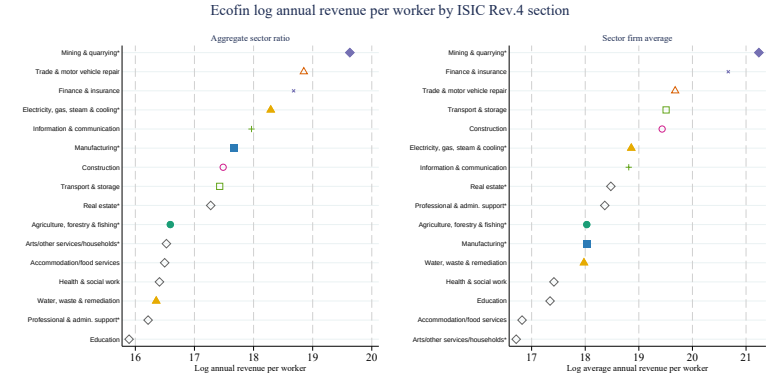


Ranks 57-84; each panel ranks divisions separately.

Appendix: same figure in levels

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Ecofin Log Revenue per Worker by ISIC Rev.4 Section



ISIC Rev.4 sections based on the NACAM-ISIC Rev.4 crosswalk.
Combined: professional & admin. support; arts/recreation, other services & household employers (division 97).
* Includes branches spanning sections; see classification notes.

Appendix: same figure in levels

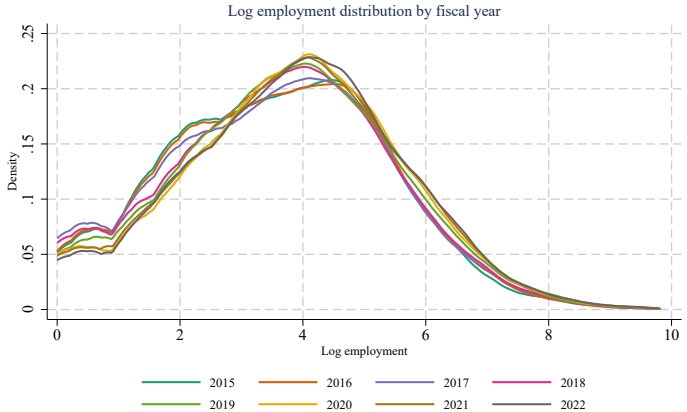
Source: Cameroon administrative Ecofin panel; calculations by authors.

Specification and Sample

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s \left(\ln(X_{it}) \times D_{ist}\right) + \varepsilon_{it} \tag{1}$$

- The unit of observation is a firm-year (i, t).
- E_{it} is employment for firm i in year t.
- X_{it} is either value added or total revenue, estimated in separate regressions.
- D_{ist} equals 1 if firm i belongs to ISIC section s in year t, and 0 otherwise.
- Group membership follows the reported classification in each firm-year; firms may change groups over time.
- α_i are firm fixed effects; δ_{st} are section–by-fiscal-year fixed effects.
- β_s is the elasticity for firms in sector s, identified through the interaction between $\ln(X_{it})$ and the sector indicator D_{ist}.
- No additional firm-level controls are included beyond these fixed effects.
- Standard errors are clustered at the firm level; reported sectors have at least 30 observations and 10 firms in both outcome samples.

Employment Distribution by Fiscal Year



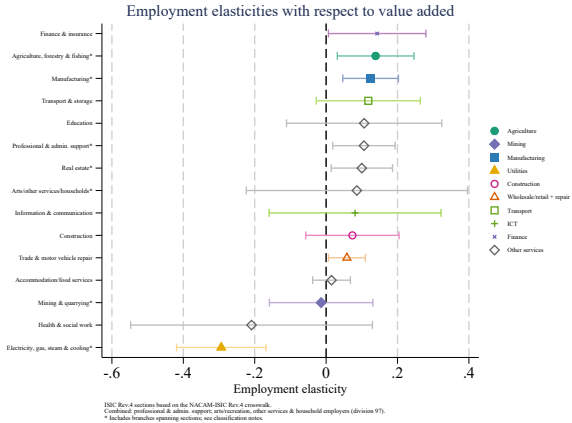
Source: Cameroon administrative Ecofin panel; calculations by authors.

Value-Added Elasticity Specification

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s^{VA} \left(\ln(VA_{it}) \times D_{ist}\right) + \varepsilon_{it} \tag{2}$$

- Estimates use the Cameroon administrative firm-year panel.
- Firm fixed effects absorb time-invariant firm differences at the taxpayer/firm level.
- Section–by-fiscal-year fixed effects absorb annual shocks common to firms in the same section.
- No age, ownership, exporter-status, or other firm controls are included.
- The plotted point for each sector is β_s^{VA} with a firm-clustered 95 percent interval.

Value-Added Elasticities Across ISIC Rev.4 Sections



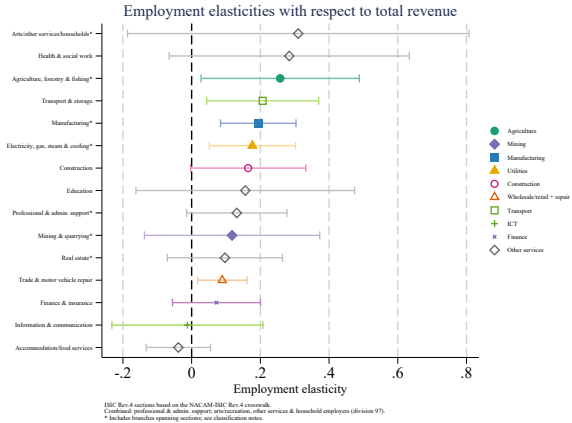
Source: Cameroon administrative Ecofin panel; calculations by authors.

Total-Revenue Elasticity Specification

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s^R \left(\ln(R_{it}) \times D_{ist}\right) + \varepsilon_{it} \tag{3}$$

- R_{it} is total revenue, estimated in a separate regression from value added.
- Fixed effects are at the firm level and section–by-fiscal-year level.
- No additional firm-level controls are included beyond these fixed effects.
- The plotted point for each sector is β_s^R with standard errors clustered at the firm level.

Total-Revenue Elasticities Across ISIC Rev.4 Sections

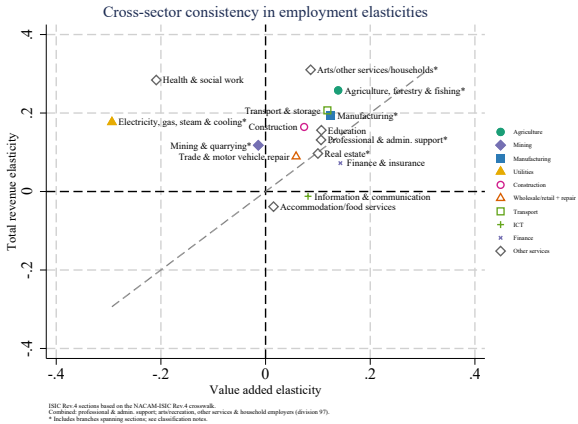


Source: Cameroon administrative Ecofin panel; calculations by authors.

Cross-Model Comparison

- The next plot compares the sector-specific value-added and total-revenue elasticities.
- Each point is one ISIC section retained by the common inclusion rule.
- Points near the 45-degree line indicate sectors where both revenue proxies give similar employment responses.
- Large gaps flag sectors where value added and gross revenue tell different stories.

Cross-Model Consistency

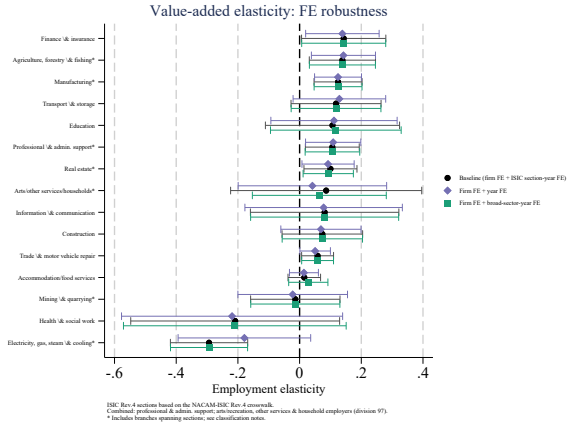


Source: Cameroon administrative Ecofin panel; calculations by authors.

Fixed-Effect Robustness Check

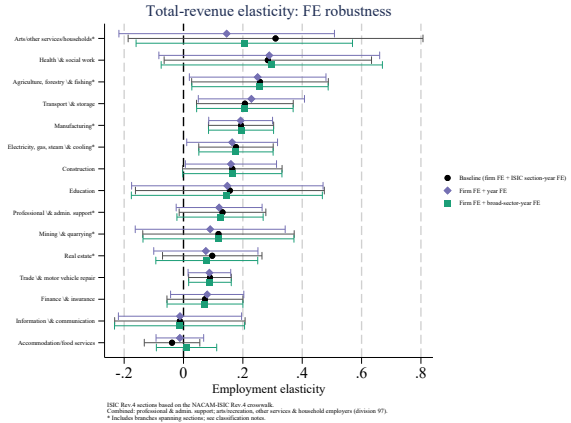
- FE means fixed effects. All robustness specifications keep taxpayer/firm fixed effects and firm-clustered standard errors.
- Baseline: firm FE plus detailed section–by-fiscal-year FE.
- Looser alternative: firm FE plus fiscal-year FE only.
- Intermediate alternative: firm FE plus broad-activity-group-by-fiscal-year FE.
- No age, ownership, exporter-status, or other firm-level controls are added in these robustness checks.

Value-Added Elasticity Robustness



Source: Cameroon administrative Ecofin panel; calculations by authors.

Total-Revenue Elasticity Robustness



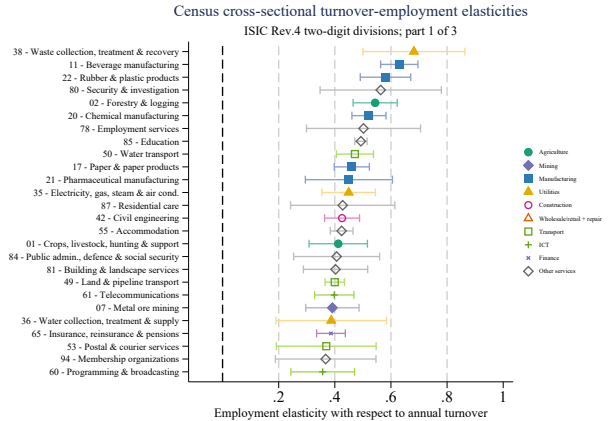
Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Turnover-Employment Specification

$$\ln(E_i) = \alpha_s + \sum_s \beta_s \left[\ln(R_i) \times D_{is}\right] + \varepsilon_i \tag{4}$$

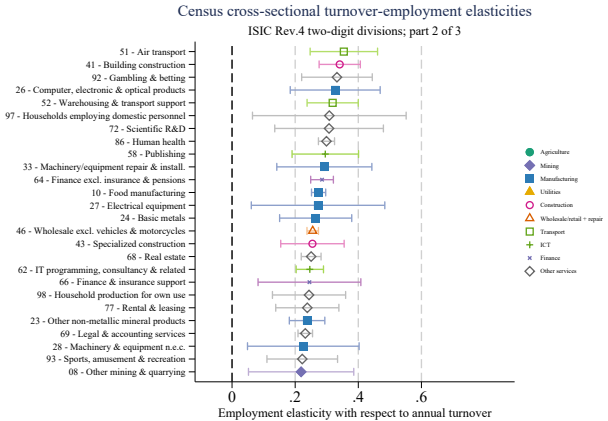
- The unit of observation is a Census headquarters firm i.
- E_i is employment and R_i is annual turnover from RGE3.
- D_{is} equals 1 if firm i belongs to ISIC Rev.4 division s.
- α_s are detailed ISIC Rev.4 division intercepts.
- β_s is the cross-sectional employment elasticity with respect to annual turnover in sector s.
- There are no firm fixed effects, year fixed effects, or division–year fixed effects because the Census has no panel time variation.
- Standard errors are heteroskedasticity-robust; reported sectors have at least 30 positive-employment and positive-turnover headquarters firms.

Census Turnover-Employment Elasticities (1/3)



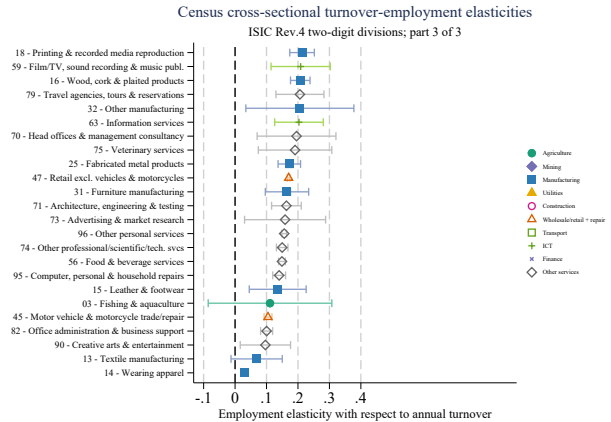
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Turnover-Employment Elasticities (2/3)



Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Turnover-Employment Elasticities (3/3)



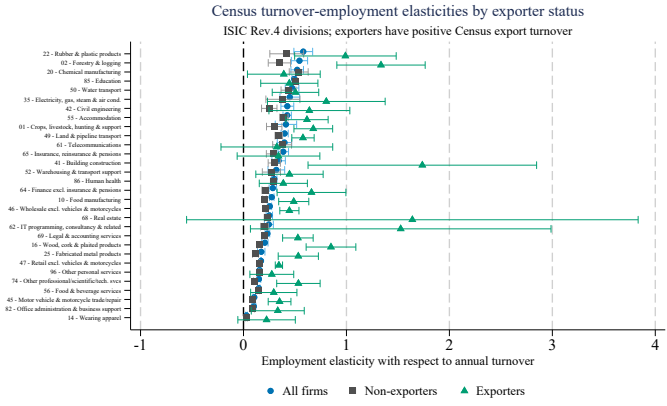
Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Exporter-Interaction Specification

$$\ln(E_i) = \alpha_s + \beta_s \ln(R_i) + \gamma_s Exporter_i + \delta_s \left[\ln(R_i) \times Exporter_i\right] + \varepsilon_i, \quad i \in s \tag{5}$$

- The unit of observation is a Census headquarters firm i in ISIC Rev.4 division s.
- E_i is employment and R_i is annual turnover from RGE3.
- $Exporter_i$ equals 1 when embedded Census export turnover is positive.
- β_s is the non-exporter elasticity; $\beta_s + \delta_s$ is the exporter elasticity.
- The total elasticity is estimated separately among all firms in the same reported sector sample.
- Standard errors are heteroskedasticity-robust.
- Reported sectors have at least 30 usable firms, 10 exporters, and 10 non-exporters.

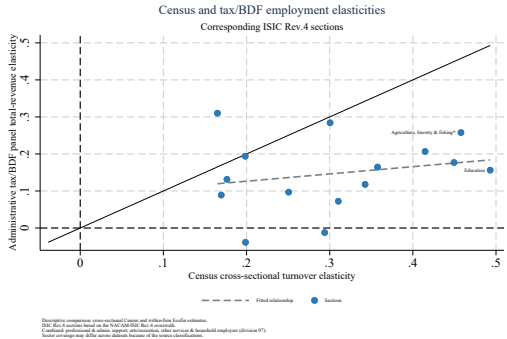
Census Turnover Elasticities by Exporter Status



Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census–Ecofin Section-Level Comparison

- Census slopes are re-estimated at the section level; Ecofin estimates use firm and section–year fixed effects.
- The lines show equality and the fitted relationship across overlapping groups. This is a descriptive comparison of different populations and sources of variation.

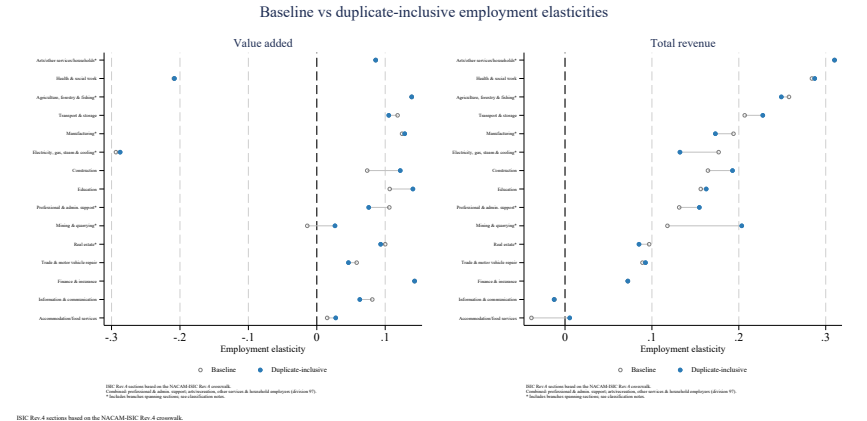


Sources: INS Cameroon, RGE3 (2023-2024), and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Duplicate Robustness Check

- The baseline specification excludes conflicting duplicate firm-years from the cleaned panel.
- The robustness specification keeps those duplicate rows as separate observations.
- Firm fixed effects and clustered standard errors still use the original firm identifier, so duplicate rows do not become new firms.
- The next plot compares baseline and duplicate-inclusive elasticities sector by sector.

Baseline vs Duplicate-Inclusive Elasticities



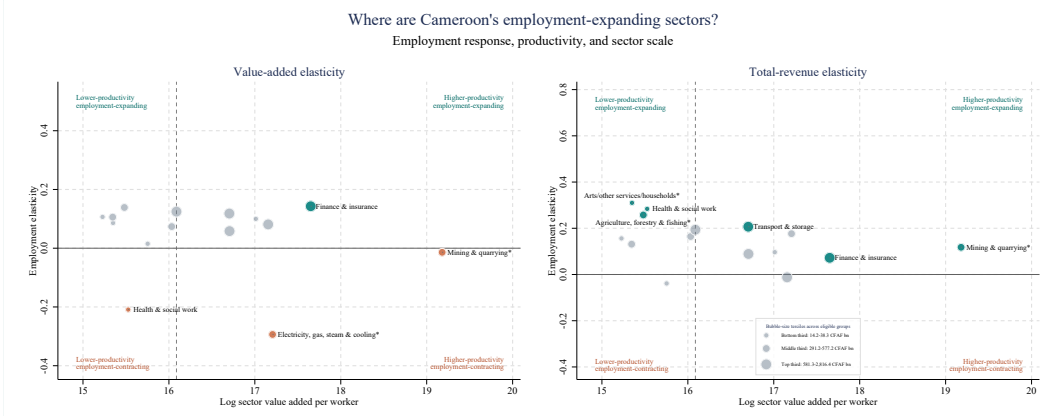
ISIC Rev 4 sections based on the NACAM-ISIC Rev 4 crosswalk.
Combined: professional & admin. support; arts/recreation, other services & household employers (division 97).
* Includes branches spanning sections; see classification notes.

Source: Cameroon administrative Ecofin panel; calculations by authors.

Elasticity, Productivity, and Sector Scale

- The opportunity maps compare sector employment elasticities with log sector value added per worker.
- The horizontal line marks zero estimated elasticity; the vertical line marks median productivity.
- Bubble size identifies the lower, middle, or upper tercile of each scale measure across reported sections; legends report the ranges.
- Labels identify extreme elasticities and the highest-productivity groups.

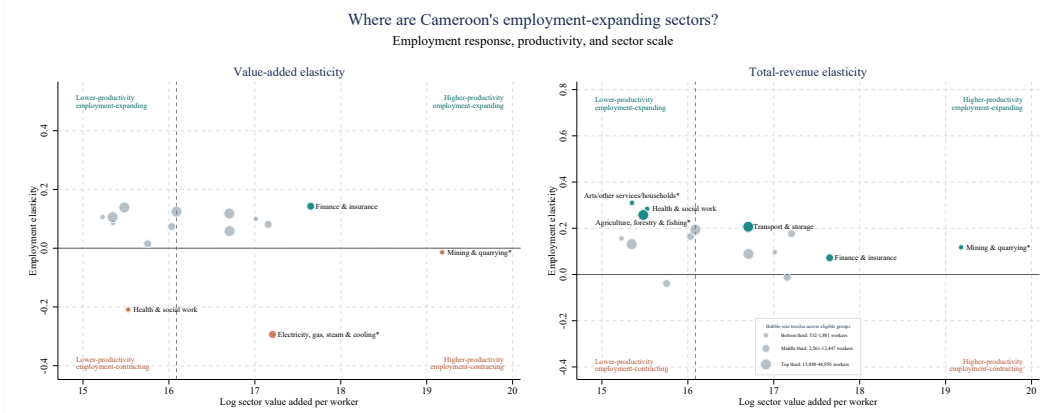
Opportunity Map: Total Sector Revenue



ISIC Rev.4 sections based on the NACAM-ISIC Rev.4 crosswalk.
Combined: professional & admin. support; arts/recreation, other services & household employers (division 97).
* Includes branches spanning sections; see classification notes.
Vertical line marks median log sector value added per worker.

Source: Cameroon administrative Ecofin panel; calculations by authors.

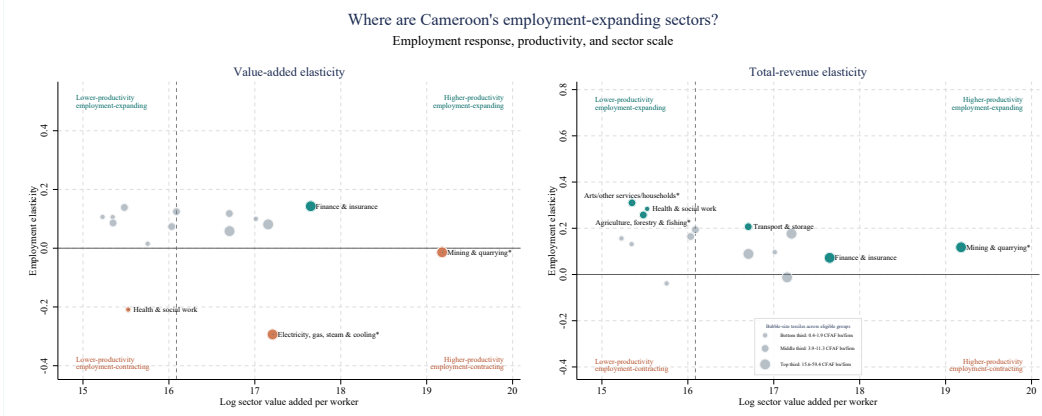
Opportunity Map: Total Sector Employment



ISIC Rev.4 sections based on the NACAM-ISIC Rev.4 crosswalk.
Combined: professional & admin. support; arts/recreation, other services & household employers (division 97).
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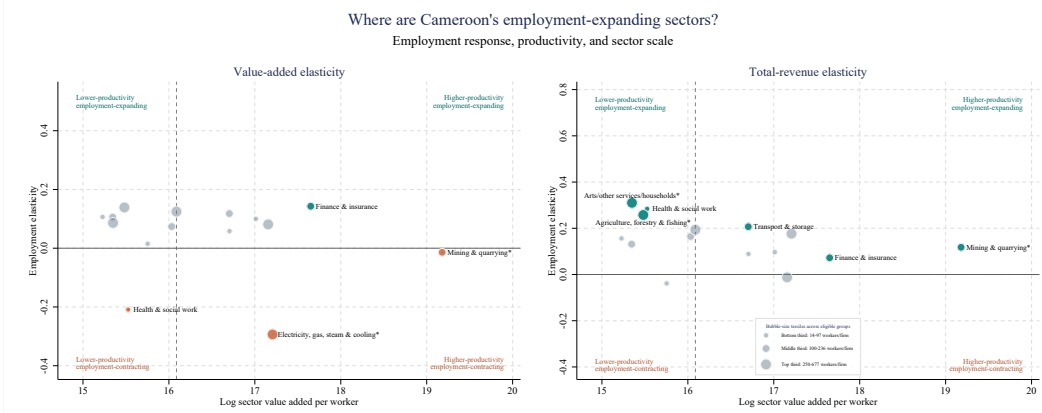
Opportunity Map: Average Firm Revenue



ISIC Rev.4 sections based on the NACAM-ISIC Rev.4 crosswalk.
Combined: professional & admin. support; arts/recreation, other services & household employers (division 97).
* Includes branches spanning sections; see classification notes.
Vertical line marks median log sector value added per worker.

Source: Cameroon administrative Ecofin panel; calculations by authors.

Opportunity Map: Average Firm Employment



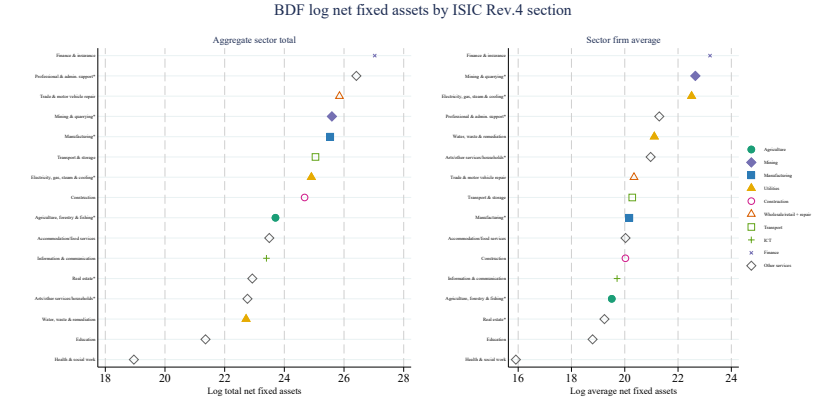
ISIC Rev.4 sections based on the NACAM-ISIC Rev.4 crosswalk.
Combined: professional & admin. support; arts/recreation, other services & household employers (division 97).
* Includes branches spanning sections; see classification notes.
Vertical line marks median log sector value added per worker.

Source: Cameroon administrative Ecofin panel; calculations by authors.

Administrative Ecofin Asset Diagnostics

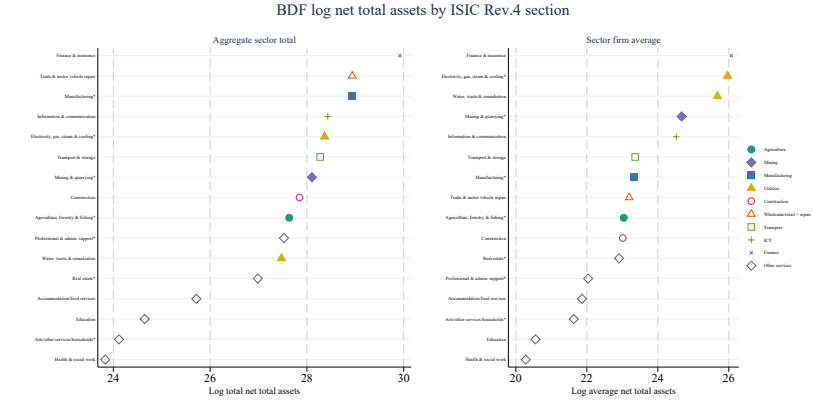
- The RGE/Census file does not include the dictionary-listed asset field, so asset diagnostics use the administrative Ecofin panel instead.
- Net fixed assets are the preferred capital proxy because they most closely represent firm capital stock in the current BDF variables.
- Each firm contributes once to the sector totals, using its latest available positive asset observation.
- Net total assets are shown as a broader balance-sheet comparison.

BDF Net Fixed Assets by ISIC Rev.4 Section



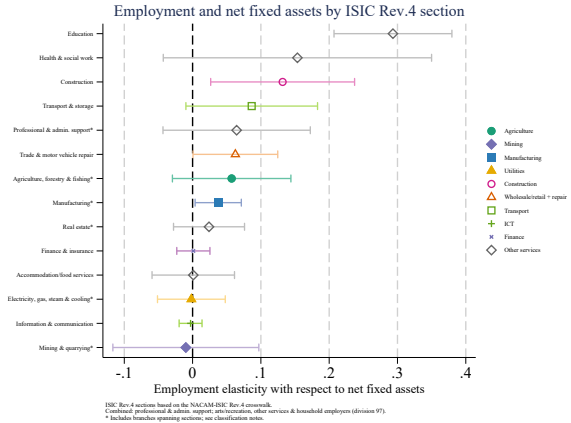
Source: Cameroon administrative Ecofin panel; calculations by authors.

BDF Net Total Assets by ISIC Rev.4 Section



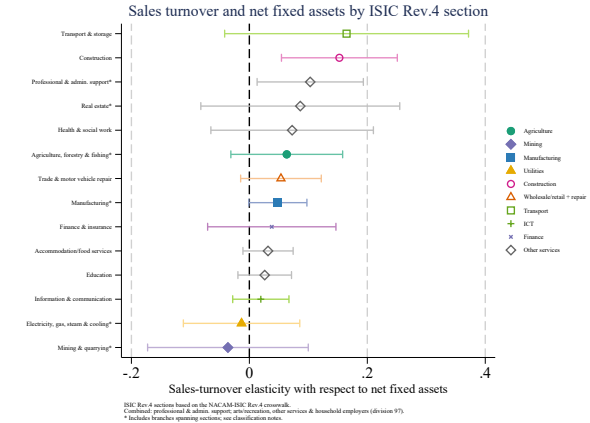
Source: Cameroon administrative Ecofin panel; calculations by authors.

Employment Response to Net Fixed Assets



Source: Cameroon administrative Ecofin panel; calculations by authors.

Sales-Turnover Response to Net Fixed Assets



Source: Cameroon administrative Ecofin panel; calculations by authors.

Revenue-Elasticity Sector Deciles

	Revenue decile	Total revenue elasticity	Value added elasticity	Revenue observations	VA observations
Arts/other services/households*	10	0.310	0.086	60	55
Health & social work	9	0.284	-0.209	263	243
Agriculture, forestry & fishing*	9	0.258	0.139	541	449
Transport & storage	8	0.207	0.118	788	708
Manufacturing*	7	0.194	0.125	1,670	1,433
Electricity, gas, steam & cooling*	7	0.177	-0.294	56	54
Construction	6	0.164	0.074	654	575
Education	5	0.156	0.107	368	364
Professional & admin. support*	5	0.131	0.106	1,387	1,264
Mining & quarrying*	4	0.118	-0.014	138	119
Real estate*	3	0.097	0.100	325	301
Trade & motor vehicle repair	3	0.089	0.058	1,712	1,522
Finance & insurance	2	0.072	0.143	300	272
Information & communication	1	-0.012	0.081	281	258
Accommodation/food services	1	-0.039	0.015	241	224

Decile 10 is highest; decile 1 is lowest.

ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Combined: professional & admin. support; arts/recreation, other services & household employers (division 97). * Includes branches spanning sections; see classification notes.

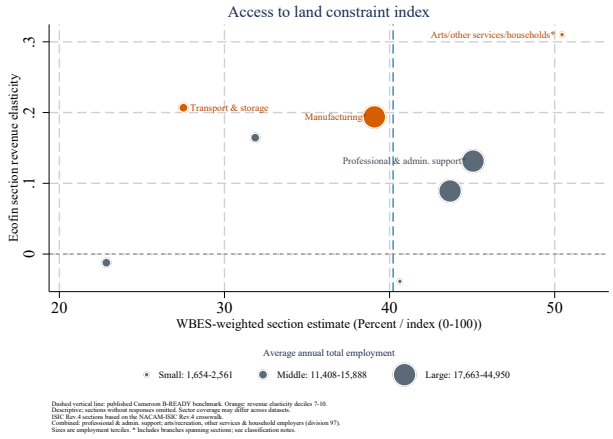
Main Takeaways

- Elasticities vary meaningfully across sectors rather than lining up around a single Cameroon-wide coefficient.
- Interpret sector differences alongside their confidence intervals and sample support; these estimates describe associations.

B-READY/WBES Constraint Extension

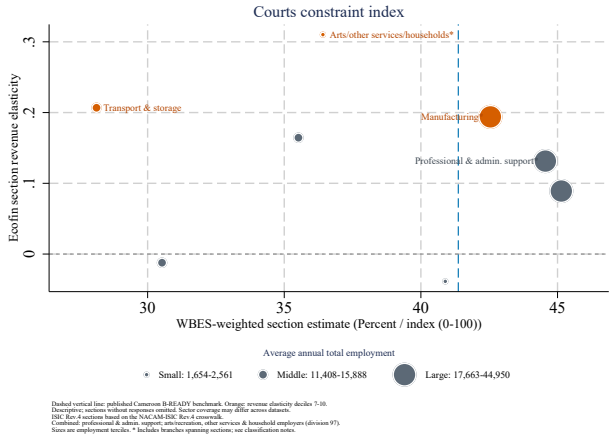
- The analysis uses the 16 questions marked `Priority = yes` in the reviewed B-READY workbook.
- Cameroon WBES firms are grouped from their ISIC Rev.4 codes into the sections used for the Ecofin comparison, combining professional & admin. support, and arts/recreation, other services & household employers (division 97).
- The horizontal axis shows WBES-weighted sector constraints; the vertical axis shows administrative total-revenue employment elasticities.
- Three bubble sizes represent employment terciles based on average annual Ecofin group employment, held fixed across indicators.
- Orange groups belong to revenue-elasticity deciles 7–10. The vertical benchmark line is independent of this highlighting rule.
- Labels identify opportunity sectors and the two largest-employment sectors with valid indicator responses.
- Results are descriptive, not causal; sectors without indicator responses are omitted from the corresponding chart.
- Fourteen indicators reproduce the national B-READY value within two points. The selected market-concentration item is unavailable in Cameroon microdata, and the VAT-refund denominator requires further review.

Access to Land Constraints by Sector



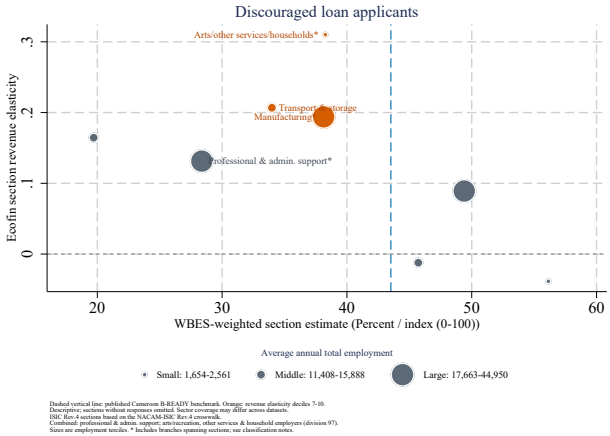
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Courts Constraints by Sector



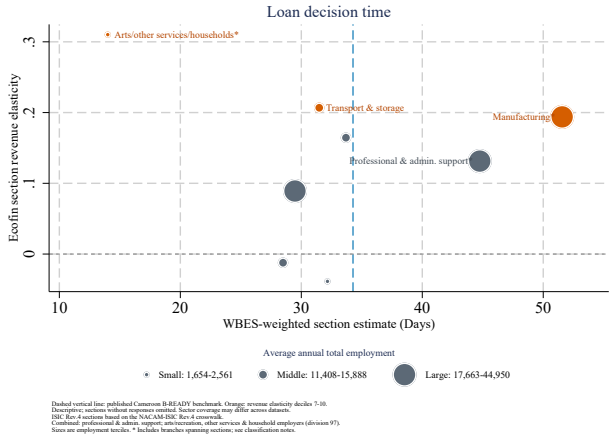
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Discouraged Loan Applicants by Sector



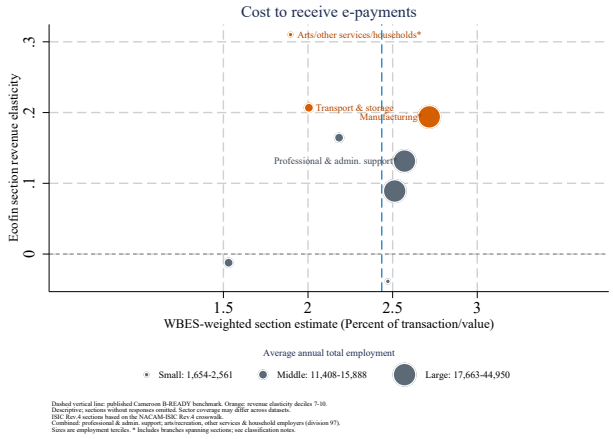
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Loan Decision Time by Sector



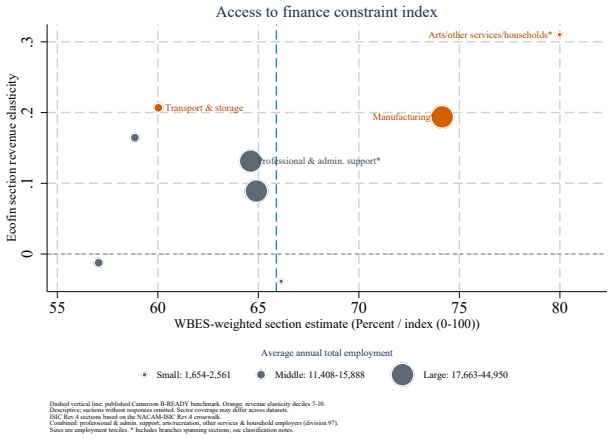
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Cost to Receive E-Payments by Sector



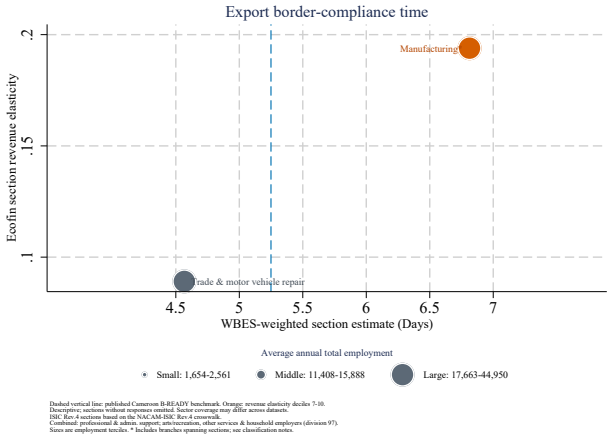
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Access to Finance Constraints by Sector



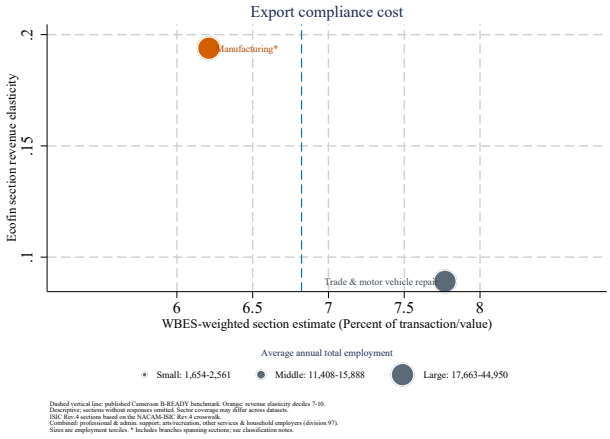
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Export Border-Compliance Time by Sector



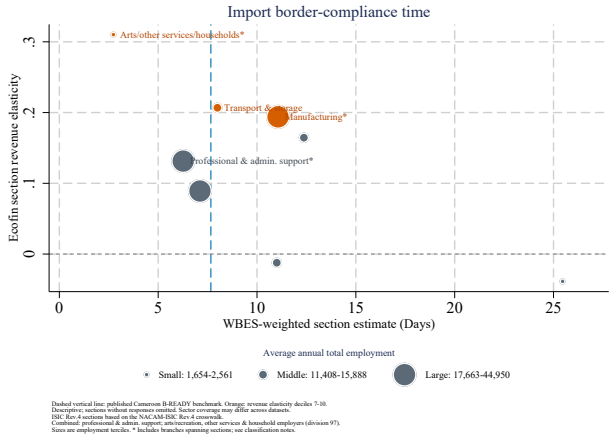
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Export Compliance Cost by Sector



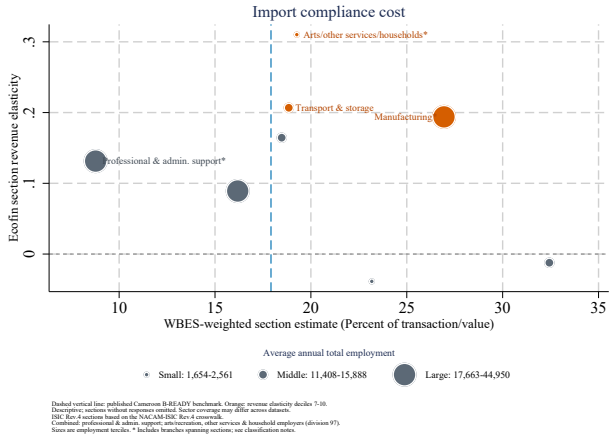
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Import Border-Compliance Time by Sector



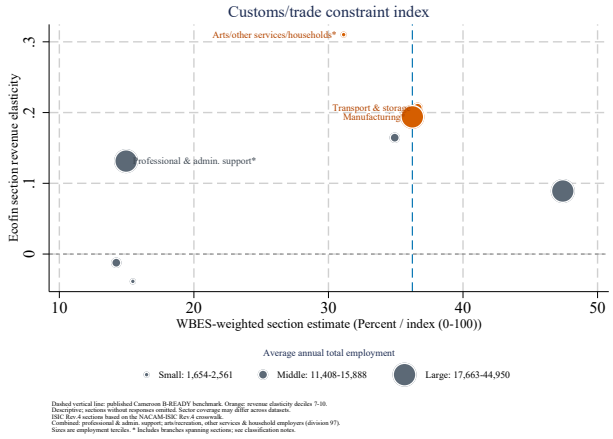
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Import Compliance Cost by Sector



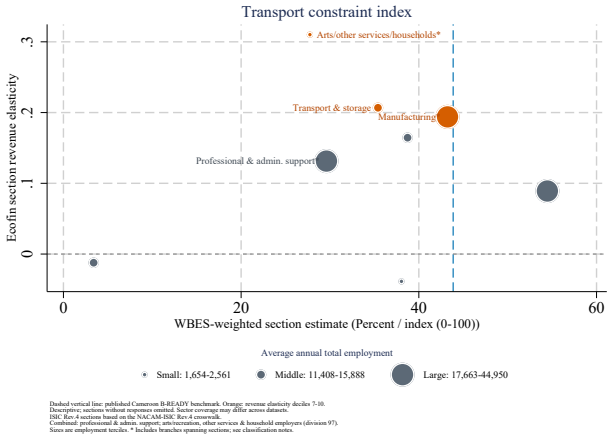
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Customs and Trade Constraints by Sector



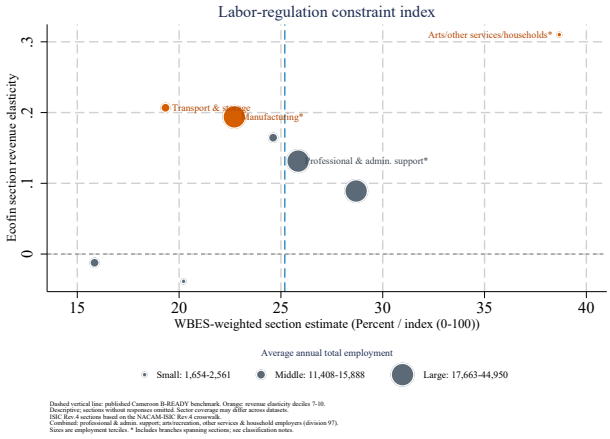
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Transport Constraints by Sector



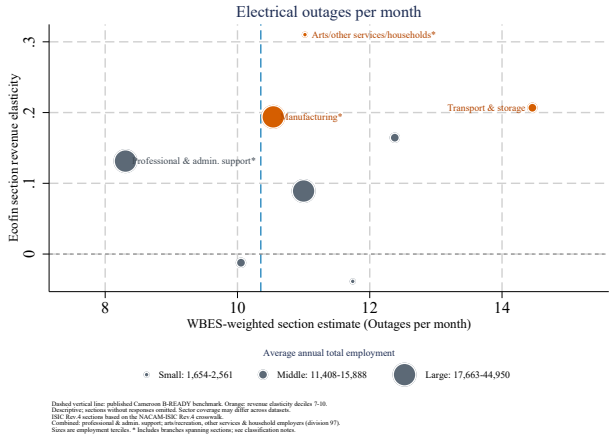
Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Labor-Regulation Constraints by Sector



Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Electrical Outages by Sector



Sources: Cameroon WBES 2024 and Ecofin. ISIC Rev.4 sections based on the NACAM–ISIC Rev.4 crosswalk. Sector coverage may differ across datasets.

Phase 3: WBES Trade Diagnostics

- Phase 3 uses the latest available World Bank Enterprise Survey wave for CEMAC countries.
- The first step is a cleaning and availability audit for sales, employment, sector, export, import, and input-origin variables.
- Gabon is not retained in the main descriptive panels because its latest available WBES wave is 2009 and has no ISIC Rev.4 coverage.
- The WBES formal-sector sampling frame covers non-agricultural private firms, so sector estimates cover the eligible non-agricultural divisions.

WBES Descriptive Statistics

	Cameroon	Central Afr. Rep.	Chad	Congo	Eq. Guinea	SSA excl. CEMAC	High-income countries
Sample composition							
Fiscal year	2023	2022	2022	2023	2023	2022	2023
Surveyed firms	615	151	164	365	173	18252	30855
Firms represented (000s)	5	5	1	2	1	354	6879
% Exporting firms	14.6	28.9	5.3	12.9	15.8	11.9	18.8
	[2.1]	[6.9]	[2.5]	[2.8]	[3.6]	[0.6]	[0.6]
% Manufacturing	17.7	15.2	28.9	29.8	43.3	24.6	16.3
	[2.7]	[5.1]	[4.4]	[3.6]	[4.5]	[0.8]	[0.5]
% Construction/utilities	8.8	15.0	3.1	4.0	7.8	10.0	15.1
	[1.7]	[5.5]	[2.3]	[1.3]	[2.7]	[0.7]	[0.6]
% Trade/hospitality/transport	61.7	67.3	59.5	61.4	34.0	60.9	53.3
	[3.3]	[6.8]	[5.3]	[3.9]	[4.6]	[1.0]	[0.8]
% Other services	11.9	2.5	8.5	4.8	14.8	4.5	15.4
	[2.0]	[1.0]	[3.5]	[1.9]	[3.2]	[0.4]	[0.7]
Economic activity (% total sales)							
Manufacturing	17.2	15.6	16.7	11.8	27.6	31.6	24.4
	[3.2]	[5.1]	[5.7]	[3.9]	[6.6]	[2.6]	[1.2]
Construction/utilities	10.5	23.0	3.5	5.9	25.8	16.1	14.1
	[3.3]	[11.6]	[2.4]	[2.8]	[9.4]	[2.4]	[1.4]
Trade/hospitality/transport	59.7	45.9	70.1	78.6	35.1	48.8	45.7
	[5.5]	[10.0]	[7.8]	[6.2]	[8.9]	[2.7]	[1.8]
Other services	12.6	15.4	9.7	3.7	11.5	3.6	15.8
	[4.4]	[10.8]	[4.4]	[3.4]	[6.2]	[0.7]	[1.5]
Average sales (current USD thousands)							
All firms	447.3	36.0	312.5	1058.7	1075.0	763.3	6325.6
	[44.3]	[6.9]	[53.1]	[226.4]	[184.5]	[38.3]	[209.6]
Exporting firms	795.4	47.1	894.1	3588.8	1002.5	1771.2	12779.7
	[166.3]	[19.1]	[444.0]	[1439.2]	[522.5]	[164.6]	[631.7]
Non-exporting firms	387.4	31.9	280.2	726.7	1088.0	633.6	4801.6
	[43.6]	[7.3]	[41.5]	[164.6]	[197.2]	[36.8]	[208.5]
Average employment							
All firms	27.8	22.3	15.2	40.9	32.0	44.9	39.2
	[2.8]	[5.3]	[1.6]	[9.0]	[4.4]	[4.9]	[4.3]
Exporting firms	51.3	14.7	23.0	151.5	19.4	93.5	73.8
	[7.3]	[2.9]	[6.6]	[57.0]	[7.4]	[19.5]	[13.8]
Non-exporting firms	23.8	24.5	14.7	26.4	34.4	38.9	31.1
	[3.1]	[8.3]	[1.7]	[4.7]	[4.9]	[5.1]	[4.2]

Sales use fiscal-year official exchange rates. Brackets report standard errors.

WBES Country Coverage

	Firms	SurveyYear	FiscalYear	FirmsWithISIC4	Retained
Cameroon	615	2,024	2,023	615	1
Central African Republic	151	2,023	2,022	151	1
Chad	164	2,023	2,022	164	1
Congo	365	2,024	2,023	365	1
Equatorial Guinea	173	2,024	2,023	173	1
Gabon	179	2,009	.	0	0

Retained excludes Gabon from the main descriptive panels because its latest available wave is 2009 and lacks ISIC Rev.4.

Core Variable Availability

	Firms	Sales	SalesW	Employment	Weight	ISIC4	ExportShare	ImportStatus	ImportValue	FirmAge	OwnershipControls	SalesShareOff100	InputShareOff100	OwnershipShareOff100
Cameroon	615	614	614	615	615	615	615	613	275	614	614	0	0	0
Central African Republic	151	139	139	151	151	151	134	126	56	147	128	0	0	0
Chad	164	164	164	164	164	164	164	164	110	164	164	0	0	0
Congo	365	352	352	358	365	365	353	331	123	363	360	0	0	0
Equatorial Guinea	173	172	172	173	173	173	173	173	108	173	173	0	0	0

Counts are unweighted establishments in retained CEMAC latest-wave surveys.

Negative Special-Code Audit

	Employment	PermanentEmp	Sales	DomesticShare	IndirectExport	DirectExport	DomesticInput	ForeignInput	DirectImport	RawMatCost	ResaleCost
Cameroon	0	0	1	0	0	0	2	2	1	12	1
Central African Republic	0	0	12	11	16	14	25	25	19	3	0
Chad	0	0	0	0	0	0	0	0	0	0	0
Congo	0	7	13	6	10	10	34	34	1	27	42
Equatorial Guinea	0	0	1	0	0	0	0	0	0	0	0

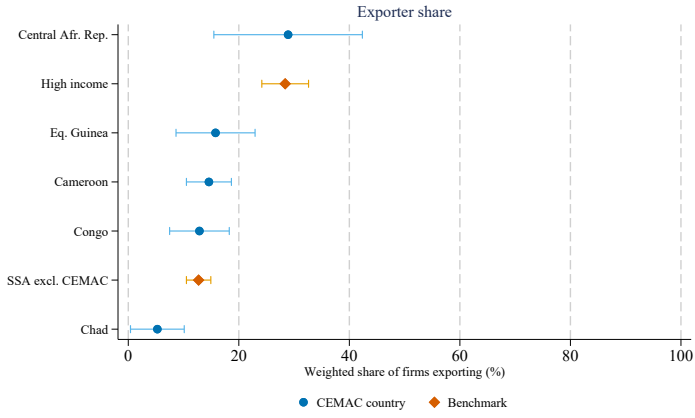
Negative WBES special codes are audited before being recoded to missing for nonnegative levels, percentages, and binary responses.

Winsorization Cutoffs

	SalesP5	SalesP95	InputCostP5	InputCostP95
Cameroon	9,125,300	2930040000	2,000,000	2800000000
Central African Republic	1,744,000	435,000,000	300,000	50,000,000
Chad	6,750,000	1115225000	1,380,000	500,000,000
Congo	1,000,000	12000000000	150,000	600,000,000
Equatorial Guinea	13,000,000	9090000000	300,000	669,000,000

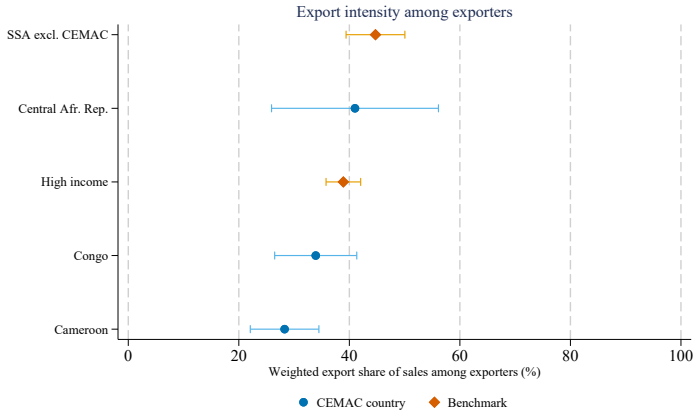
Sales and input costs are winsorized at the 5th and 95th percentiles within each country wave.

Exporter Participation



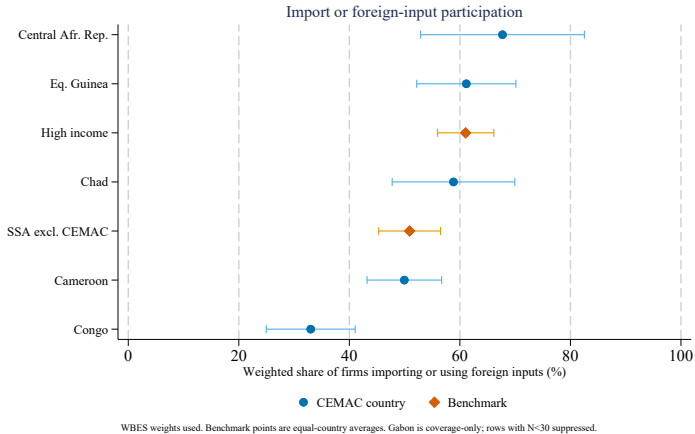
Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Export Intensity Among Exporters



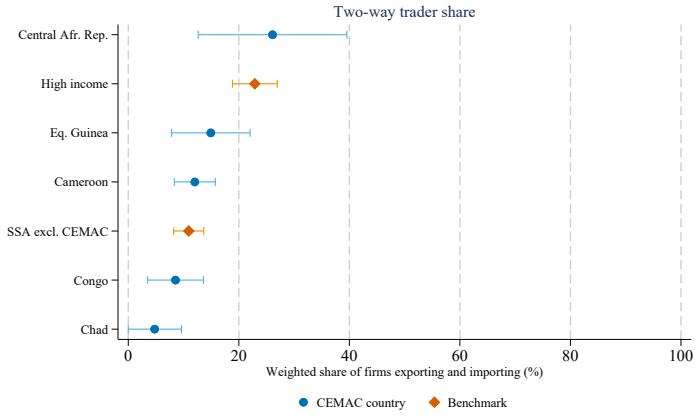
Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Import and Foreign-Input Participation



Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Two-Way Trader Status



Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Focal WBES Revenue-Exporter Specification

$$\ln(E_i) = \alpha_c + \beta_c \ln(R_i) + \gamma_c \mathit{Exporter}_i + \delta_c \left[\ln(R_i) \times \mathit{Exporter}_i\right] + \theta_c Z_i + \lambda_{cq(i)} + \varepsilon_i, \quad i \in c \quad (6)$$

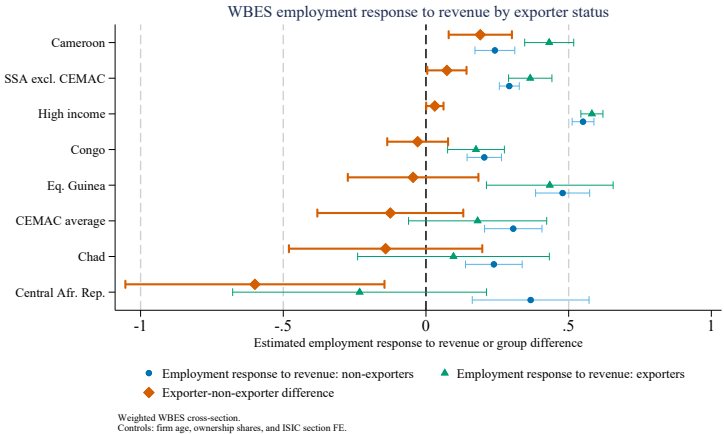
- c indexes the country. The model is estimated separately within each country.
- WBES is cross-sectional, so there are no firm fixed effects or panel controls.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{cq(i)}$ are country-specific ISIC Rev.4 section fixed effects.
- Estimates use WBES weights and robust standard errors, not clustered standard errors.
- β_c is the non-exporter slope; $\beta_c + \delta_c$ is the exporter slope.
- The focal estimate is δ_c, the exporter minus non-exporter slope difference.
- Display rule: at least 30 firms overall and at least 10 firms in each exporter group.

Country Revenue-Exporter Interaction Estimates

	Non-exporter	Exporter	Difference	SE	Lower 95	Upper 95	P-value	N	Exporters	Non-exporters
Cameroon	0.241	0.432	0.190	0.056	0.080	0.301	0.001	612	136	476
Central Afr. Rep.	0.367	-0.232	-0.599	0.229	-1.053	-0.145	0.010	108	36	72
Chad	0.238	0.096	-0.141	0.171	-0.480	0.197	0.411	164	13	151
Congo	0.204	0.175	-0.029	0.054	-0.136	0.077	0.591	334	46	288
Eq. Guinea	0.479	0.434	-0.045	0.116	-0.274	0.184	0.698	172	18	154
CEMAC average	0.306	0.181	-0.125	0.130	-0.381	0.131	0.338	1,390	249	1,141
High income	0.550	0.581	0.031	0.015	0.001	0.061	0.045	28,692	10,697	17,995
SSA excl. CEMAC	0.292	0.365	0.073	0.035	0.005	0.142	0.036	14,990	2,693	12,297

The difference column is the focal exporter minus non-exporter slope estimate.

Country Employment Responses to Revenue by Exporter Status



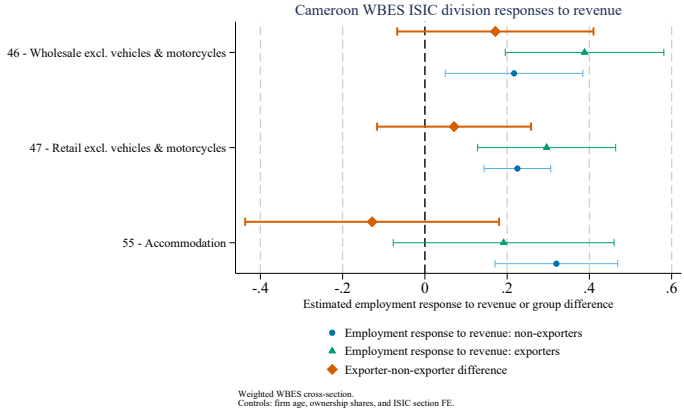
Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Cameroon WBES Division Specification

$$\ln(E_i) = \alpha_a + \beta_a \ln(R_i) + \gamma_a Exporter_i + \delta_a [\ln(R_i) \times Exporter_i] + \theta_a Z_i + \varepsilon_i, \quad i \in a \tag{7}$$

- a indexes two-digit ISIC Rev.4 divisions; the model is estimated separately within each division.
- WBES is cross-sectional, so there are no firm fixed effects.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- Each division belongs to one section, so section controls are absorbed by its intercept.
- Estimates use WBES weights and robust standard errors, not clustered standard errors.
- Reported divisions require at least 30 usable firms, including 10 exporters and 10 non-exporters; excluded divisions remain in the sample audit.

Cameroon ISIC Division Employment Responses to Revenue (Diagnostic)



Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Division estimates are exploratory diagnostics; suppressed groups remain documented in the generated sample audit.

Supporting WBES Export-Value Specification

$$\ln(E_i) = \alpha + \beta \ln(X_i^+) + \gamma \mathit{Exporter}_i + \theta Z_i + \lambda_{q(i)} + \varepsilon_i \tag{8}$$

- These supporting models use export value rather than total revenue.
- Latest-wave WBES estimates remain weighted cross-sectional associations.
- X_i^+ is export value, with log export value set to zero for non-exporters in the all-firm model.
- The exporter dummy separates the extensive margin from the positive-export-value slope.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{q(i)}$ are ISIC Rev.4 section fixed effects.
- Rows with fewer than 30 observations are suppressed; benchmark rows are equal-country averages.

WBES Country Trade-Employment Estimates

	All-firm slope	SE	N	Exporters	Exporter-only slope	SE	N	Countries
Cameroon	0.325	0.038	613	136	0.373	0.040	136	1
Central Afr. Rep.	-0.117	0.208	112	36	-0.273	0.113	36	1
Chad	-0.044	0.176	164	13	.	.	.	1
Congo	0.115	0.045	340	46	0.118	0.068	46	1
Eq. Guinea	0.297	0.118	172	18	.	.	.	1
CEMAC average	0.115	0.088	1,401	249	0.072	0.188	218	5
High income	0.299	0.014	29,863	10,704	0.319	0.013	10,642	45
SSA excl. CEMAC	0.240	0.031	16,820	2,724	0.269	0.027	2,546	38

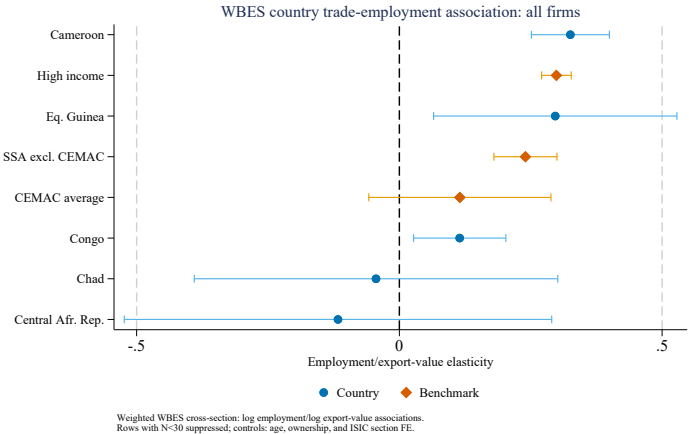
Slopes are weighted cross-sectional log employment/log export-value associations.

WBES Country Specification: All Firms

$$\ln(E_i) = \alpha_c + \beta_c \ln(X_i^+) + \gamma_c Exporter_i + \theta_c Z_i + \lambda_{cq(i)} + \varepsilon_i \tag{9}$$

- Estimated separately by country using WBES weights.
- Non-exporters stay in the sample with $\ln(X_i^+) = 0$.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{cq(i)}$ are country-specific ISIC Rev.4 section fixed effects.
- The plotted coefficient is β_c; it is descriptive, not causal.

WBES Country Associations: All Firms



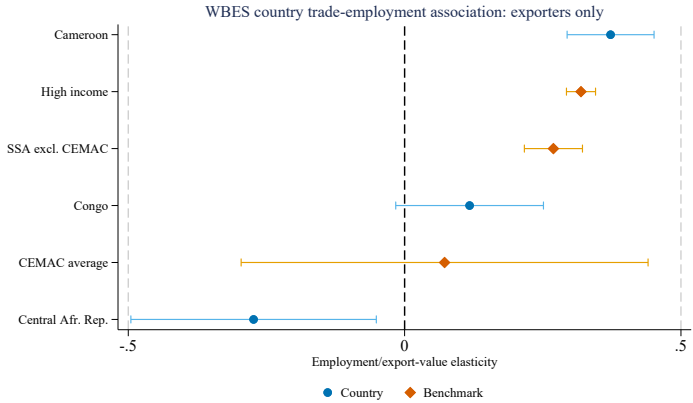
Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

WBES Country Specification: Exporters Only

$$\ln(E_i) = \alpha_c + \beta_c \ln(X_i) + \theta_c Z_i + \lambda_{cq(i)} + \varepsilon_i, \quad X_i > 0 \tag{10}$$

- Estimated separately by country among firms with positive export value.
- This is the intensive-margin association within exporters.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- $\lambda_{cq(i)}$ are country-specific ISIC Rev.4 section fixed effects.
- Countries with fewer than 30 exporters are suppressed in the plot.

WBES Country Associations: Exporters Only



Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Cameroon WBES Division Estimates

	All-firm slope	SE	N	Exporters	Exporter-only slope	SE	N
10 - Food manufacturing	0.557	0.104	48	8	.	.	.
46 - Wholesale excl. vehicles & motorcycles	0.405	0.097	82	24	.	.	.
47 - Retail excl. vehicles & motorcycles	0.317	0.052	126	19	.	.	.
55 - Accommodation	0.084	0.093	36	10	.	.	.
56 - Food & beverage services	-0.006	0.008	30	1	.	.	.

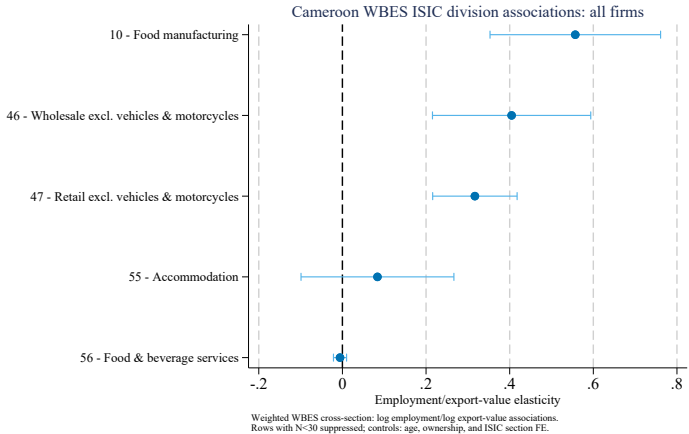
Groups are two-digit ISIC Rev.4 divisions; short English labels adapted from INSEE NAF Rev.2.

Cameroon ISIC Division Specification: All Firms

$$\ln(E_i) = \alpha_a + \beta_a \ln(X_i^+) + \gamma_a Exporter_i + \theta_a Z_i + \varepsilon_i \tag{11}$$

- Estimated within Cameroon WBES divisions.
- This is a diagnostic adaptation of the elasticity idea to cross-sectional survey data.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- Each division belongs to one section, so section controls are absorbed by its intercept.
- Divisions must meet the support rules separately for the all-firm and exporter-only samples.

Cameroon ISIC Division Associations: All Firms



Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Cameroon ISIC Division Specification: Exporters Only

$$\ln(E_i) = \alpha_a + \beta_a \ln(X_i) + \theta_a Z_i + \varepsilon_i, \quad X_i > 0 \tag{12}$$

- Estimated among Cameroon exporters within WBES divisions.
- Z_i controls for log firm age, foreign ownership share, and government/state ownership share.
- Each division belongs to one section, so section controls are absorbed by its intercept.
- Rows are shown only when at least 30 exporter observations are available.
- These results are best read as sample diagnostics for follow-up, not sector multipliers.

Cameroon ISIC Division Associations: Exporters Only

Cameroon WBES ISIC division associations: exporters only

No supported ISIC division estimates

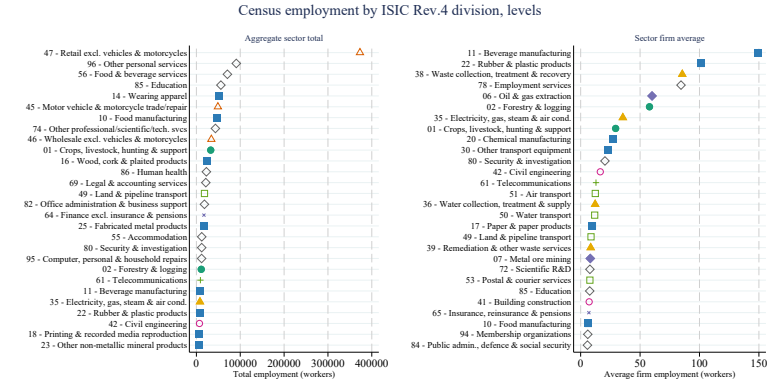
Minimum N=30; all-firm models require both export statuses.

Source: World Bank Enterprise Surveys, latest available CEMAC waves; calculations by authors.

Ecofin Sector Classification Notes

- ISIC Rev.4 sections are based on the NACAM–ISIC Rev.4 crosswalk. Asterisks identify sections containing branches that span section boundaries, including combined sections.
- Combined sections: professional & admin. support; arts/recreation, other services & household employers. Household employers cover division 97 only.
- Mining includes branches 005 and 006, whose definitions also span manufacturing and, for 005, transport.
- Agriculture includes branch 009 (cocoa, coffee, tea and sugar, including processing); manufacturing includes branch 021 (rubber and plastics, with agriculture references).
- Electricity includes branch 029, which also spans manufacturing; real estate includes branch 037, which also spans construction.
- These branch assignments retain the reviewed crosswalk decisions. Sector coverage may differ across Census, Ecofin and WBES because of the source classifications.

Census Employment by ISIC Rev.4 Division, Levels (1/3)

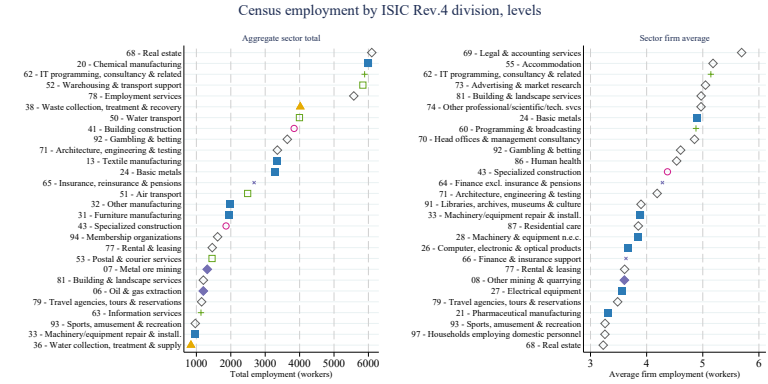


Ranks 1-28; each panel ranks divisions separately.

Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Employment by ISIC Rev.4 Division, Levels (2/3)

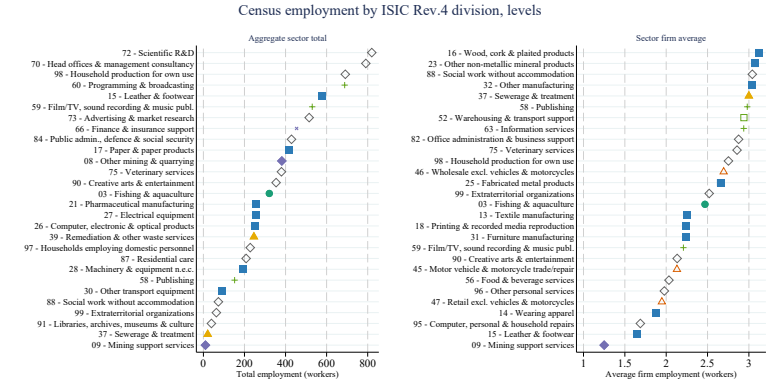


Ranks 29-56; each panel ranks divisions separately.

Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Employment by ISIC Rev.4 Division, Levels (3/3)

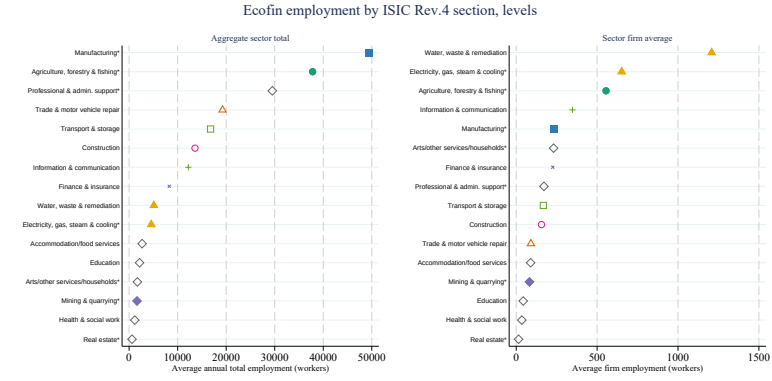


Ranks 57-84; each panel ranks divisions separately.

Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

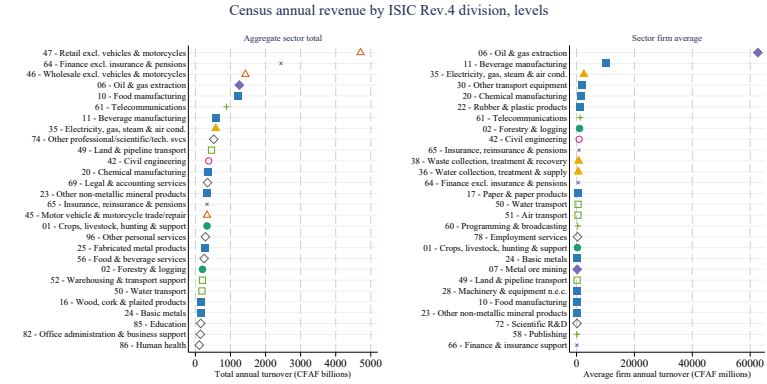
Ecofin Employment by ISIC Rev.4 Section, Levels



Back to log figure

Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Revenue by ISIC Rev.4 Division, Levels (1/3)

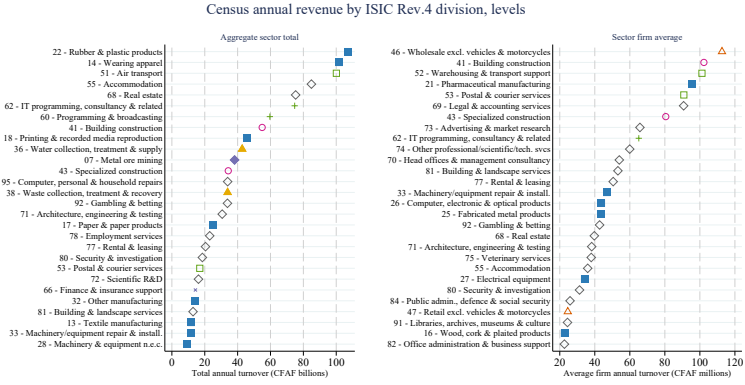


Ranks 1-28; each panel ranks divisions separately.

Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Revenue by ISIC Rev.4 Division, Levels (2/3)

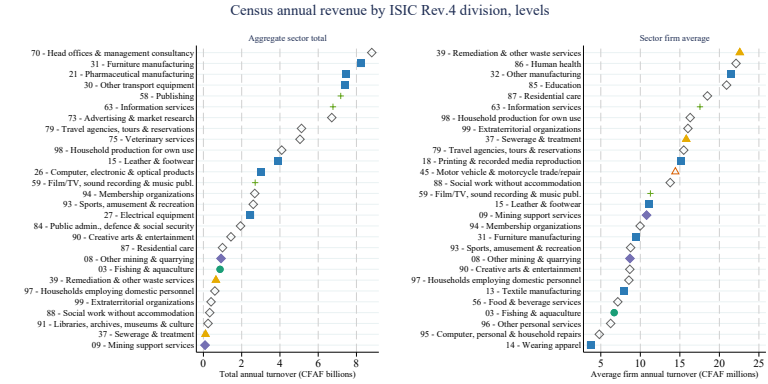


Ranks 29-56; each panel ranks divisions separately.

Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Revenue by ISIC Rev.4 Division, Levels (3/3)

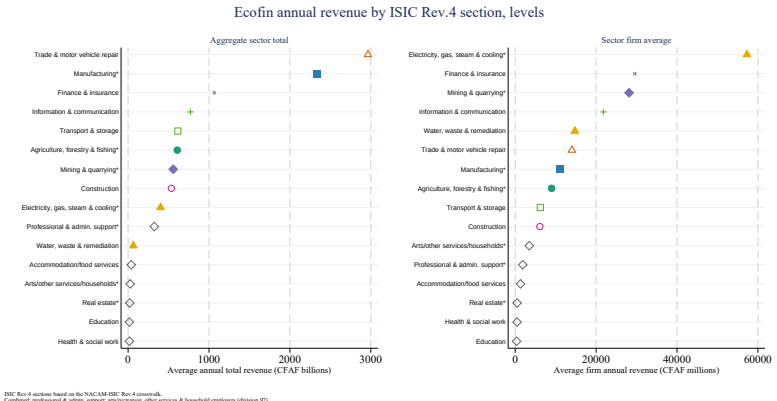


Ranks 57-84; each panel ranks divisions separately.

Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

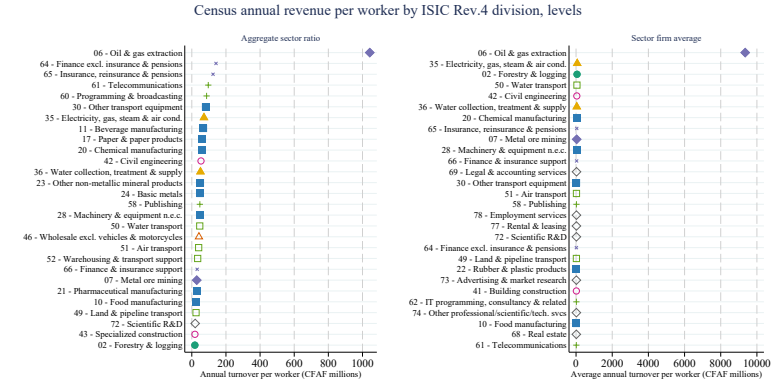
Ecofin Revenue by ISIC Rev.4 Section, Levels



Back to log figure

Source: Cameroon administrative Ecofin panel; calculations by authors.

Census Revenue per Worker by ISIC Rev.4 Division, Levels (1/3)

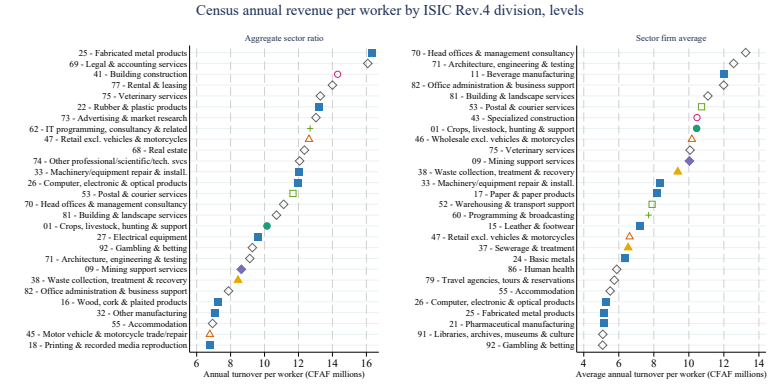


Ranks 1-28; each panel ranks divisions separately.

Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Revenue per Worker by ISIC Rev.4 Division, Levels (2/3)

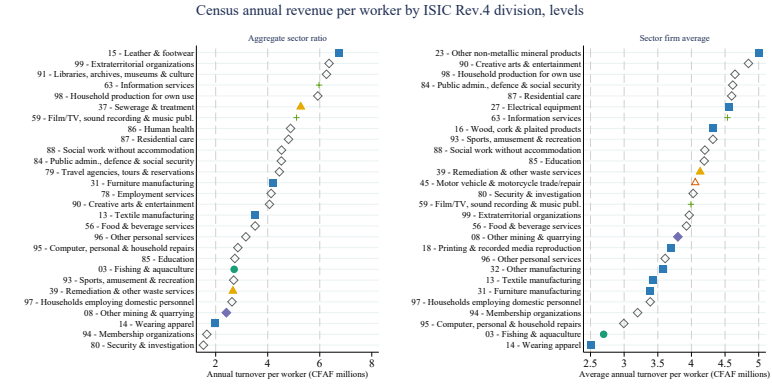


Ranks 29-56; each panel ranks divisions separately.

Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Census Revenue per Worker by ISIC Rev.4 Division, Levels (3/3)

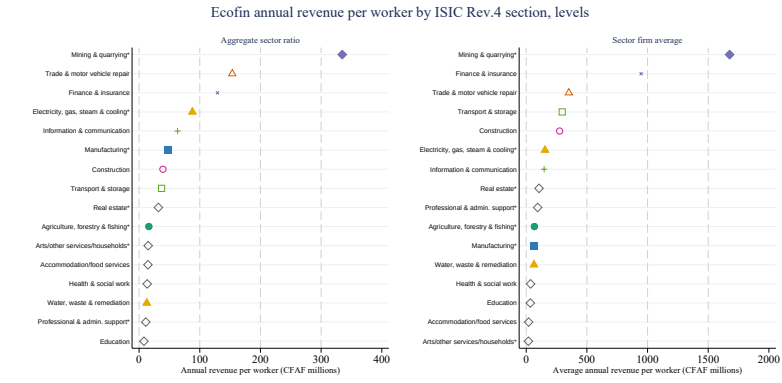


Ranks 57-84; each panel ranks divisions separately.

Back to log figure

Source: INS Cameroon, RGE3 (2023-2024); calculations by authors.

Ecofin Revenue per Worker by ISIC Rev.4 Section, Levels



Back to log figure

Source: Cameroon administrative Ecofin panel; calculations by authors.